# Trigonometric Functions & the Unit Circle

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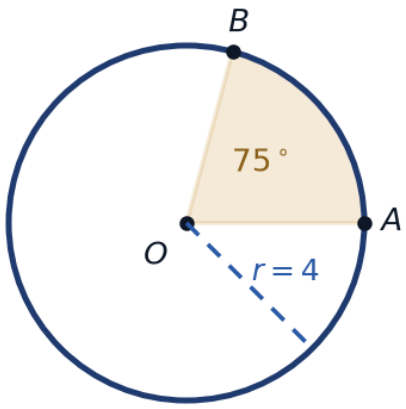
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Radian Measure & Unit Circle Coordinates

A terminal ray in standard position passes through the point $P(-8, 15)$. Find the exact values of $\sin\theta$, $\cos\theta$, and $\tan\theta$.

**2** [Foundational] • Target: Exact Values of Trigonometric Functions

Given vectors $\mathbf{u} = \langle 4, -3 \rangle$ and $\mathbf{v} = \langle -1, 5 \rangle$, compute: (a) $2\mathbf{u} - 3\mathbf{v}$, (b) the magnitude $||\mathbf{u}||$ and unit vector in the direction of $\mathbf{u}$, and (c) the dot product $\mathbf{u} \cdot \mathbf{v}$.




**3** [Foundational] • Target: Sinusoidal Graphs & Transformations

State the amplitude, period, phase shift, and midline of $f(x) = 3\sin(2x - \pi) + 1$.

**4** [Foundational] • Target: Fundamental Trigonometric Identities

A population doubles every 12 years, starting from $P_0 = 500$. When does it reach 4000?

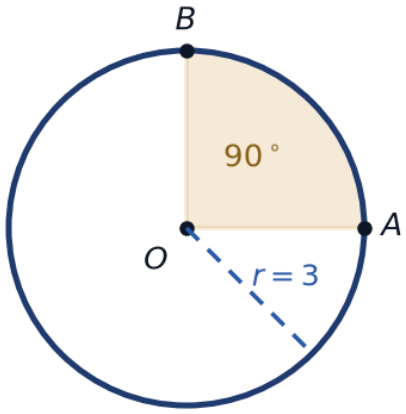
**5** [Foundational] • Target: Radian Measure & Unit Circle Coordinates

Convert the polar equation $r = 6\cos\theta - 8\sin\theta$ to Cartesian rectangular form, and identify the geometric curve, its center, and its radius.

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**6** [Foundational] • Target: Exact Values of Trigonometric Functions

Eliminate the parameter $t$ from the parametric curve $x(t) = 3 + 4\cos t$, $y(t) = -2 + 5\sin t$ for $0 \le t \le 2\pi$. Identify the conic section, its center, and its vertices.




**7** [Foundational] • Target: Sinusoidal Graphs & Transformations

A particle moves along $x(t) = 2t + 1,\ y(t) = t^2 - 3.$ Find the Cartesian equation of the path.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Fundamental Trigonometric Identities

A transformation maps point $A = (1, 2)$ to $A' = (-4, 2)$. If this transformation is a translation, what is the component form of the translation vector $\mathbf{v}$?

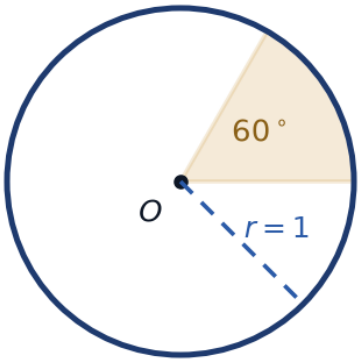
**9** [Intermediate] • Target: Radian Measure & Unit Circle Coordinates

Consider the transformation grid below. If the original point $A$ is mapped to $A'$, what is the vector representing this translation?

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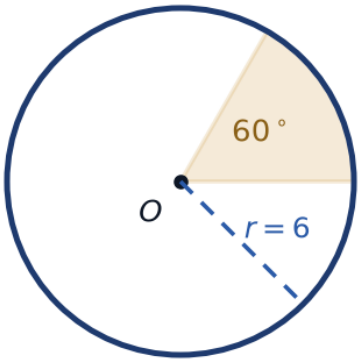
**10** **[Intermediate]** • **Target: Exact Values of Trigonometric Functions**

A point $P$ is on the unit circle such that its terminal ray makes an angle of $\theta$ with the positive x-axis. The coordinates of $P$ are $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$. If a point $Q$ on the unit circle has an angle $\phi$ such that $\phi = \theta + \frac{\pi}{3}$, what is the y-coordinate of point $Q$? Show your complete algebraic work.




**11** **[Intermediate]** • **Target: Sinusoidal Graphs & Transformations**

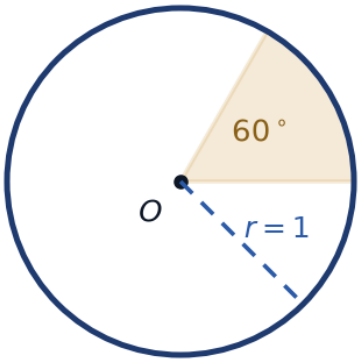
The point P is located on the unit circle at an angle of $\frac{\pi}{3}$ radians counterclockwise from the positive x-axis. What are the Cartesian coordinates $(x, y)$ of point P? Show your complete algebraic work.




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**12** **[Intermediate]** • **Target: Fundamental Trigonometric Identities**

A point $P$ is located on the unit circle such that its angle with the positive x-axis is $\frac{2\pi}{3}$ radians. What are the coordinates of point $P$? Show your complete algebraic work.

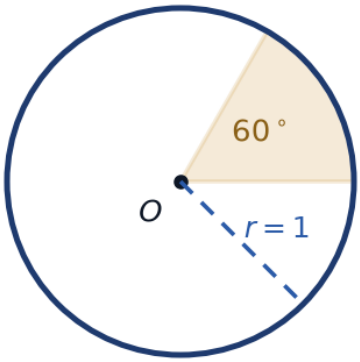



**13** **[Intermediate]** • **Target: Radian Measure & Unit Circle Coordinates**

A circle is centered at the origin, and a point P lies on the circle. The angle formed by the positive x-axis and the line segment OP (where O is the origin) is $\theta$. The coordinates of point P are $(r\cos\theta, r\sin\theta)$. If point P has coordinates $\left(\frac{5}{2}, \frac{5\sqrt{3}}{2}\right)$ and lies on a circle of radius $R$, find the value of $R$.

**14** **[Intermediate]** • **Target: Exact Values of Trigonometric Functions**

A point P lies on the unit circle such that its terminal ray makes an angle of $\theta$ with the positive x-axis. If the x-coordinate of P is $\frac{1}{2}$, what is the y-coordinate of P? Assume $\theta$ is in the first quadrant.

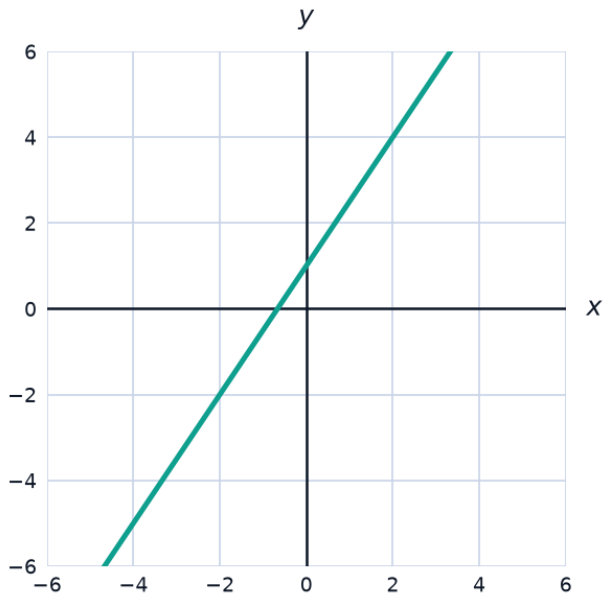



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## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

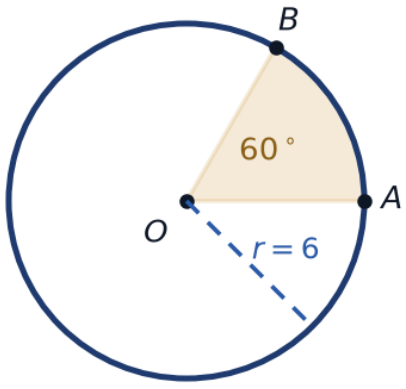
**15** [Challenging] • Target: Sinusoidal Graphs & Transformations • Cross-Lesson: 1.02 Rational Functions & Asymptotes

In polar form, which equation represents a circle of radius 3 centred at the origin? Show your complete algebraic work.




**16** [Challenging] • Target: Fundamental Trigonometric Identities • Cross-Lesson: 1.03 Rates of Change & Concavity

A particle's position in the Cartesian plane is described by the polar coordinates $(r, \theta)$, where $r$ is the distance from the origin and $\theta$ is the angle measured counterclockwise from the positive x-axis. The particle begins at $\theta_1 = \frac{\pi}{6}$ with a radial distance of $r_1 = 4$. It then moves to a new position such that its new angle is $\theta_2 = \frac{2\pi}{3}$ and the radial distance is $r_2 = 6$. If the displacement vector of the particle is $\mathbf{d} = \langle x, y \rangle$, what is the value of $x^2 + y^2$? Show your complete algebraic work.




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**17** [Challenging] • Target: Radian Measure & Unit Circle Coordinates • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Consider a particle whose position in the complex plane at time $t$ (in seconds) is given by $z(t) = 3e^{i(\frac{\pi}{6}t + \frac{\pi}{3})}$. (a) Find the initial position of the particle at $t = 0$ in polar coordinates $(r, \theta)$, where $r \geq 0$ and $0 \leq \theta < 2\pi$. (b) Determine the angular velocity of the particle in radians per second. (c) If the particle starts at a point A and moves along a circular path, find the arc length traveled by the particle from $t = 0$ to $t = 4$ seconds.

**18** [Challenging] • Target: Exact Values of Trigonometric Functions • Cross-Lesson: 2.02 Logarithmic Functions & Properties

A point P has polar coordinates $(r, \theta)$. When converted to Cartesian coordinates, P is $(3, 3\sqrt{3})$. If the angle $\theta$ is restricted to the interval $[0, 2\pi)$, find the value of $r$ and $\theta$ in radians.

**19** [Challenging] • Target: Sinusoidal Graphs & Transformations • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Write a sinusoidal function with amplitude 5, period $4\pi$, midline $y = -3$, using $\cos$.

**20** [Challenging] • Target: Fundamental Trigonometric Identities • Cross-Lesson: 4.01 Inverse & Composite Functions

Write a sinusoidal function with amplitude 5, period $3\pi$, midline $y = -2$, using $\sin$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Radian Measure & Unit Circle Coordinates

What are the exact coordinates of the terminal point on the unit circle for angle $\theta = \frac{5\pi}{6}$?

**a.** $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

**b.** $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

**c.** $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

**d.** $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

**22** [Foundational] • Target: Radian Measure & Unit Circle Coordinates

Convert an angle of $\frac{7\pi}{4}$ radians into degree measure.

**a.** $315^\circ$

**b.** $225^\circ$

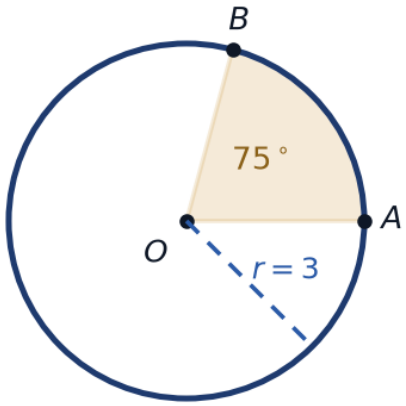
**c.** $300^\circ$

**d.** $135^\circ$

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**23** [Foundational] • Target: Radian Measure & Unit Circle Coordinates

If the terminal ray of angle $\alpha$ passes through the point $(-3, -4)$, what is $\sin\alpha$?




**a.** $-\frac{4}{5}$

**b.** $-\frac{3}{5}$

**c.** $\frac{4}{3}$

**d.** $\frac{3}{5}$

**24** [Foundational] • Target: Exact Values of Trigonometric Functions

Evaluate $\tan\left(\frac{4\pi}{3}\right)$.

**a.** $\sqrt{3}$

**b.** $-\sqrt{3}$

**c.** $\frac{\sqrt{3}}{3}$

**d.** $-\frac{\sqrt{3}}{3}$

**25** [Foundational] • Target: Exact Values of Trigonometric Functions

What is the exact value of $\sec\left(-\frac{\pi}{4}\right)$?

**a.** $\sqrt{2}$

**b.** $-\sqrt{2}$

**c.** $\frac{\sqrt{2}}{2}$

**d.** $2$

**26** [Foundational] • Target: Exact Values of Trigonometric Functions

If $\sin\theta = \frac{3}{5}$ and $\cos\theta < 0$, find $\cot\theta$.

**a.** $-\frac{4}{3}$

**b.** $\frac{4}{3}$

**c.** $-\frac{3}{4}$

**d.** $-\frac{4}{5}$

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**27** [Foundational] • Target: Sinusoidal Graphs & Transformations

Find the amplitude and period of the sinusoidal function $f(x) = -4\cos(3x - \pi) + 2$.

**a.** Amplitude $= 4$, Period $= \frac{2\pi}{3}$

**b.** Amplitude $= -4$, Period $= 3$

**c.** Amplitude $= 4$, Period $= 3\pi$

**d.** Amplitude $= 2$, Period $= \frac{2\pi}{3}$

**28** [Foundational] • Target: Sinusoidal Graphs & Transformations

A Ferris wheel of radius $25$ m completes one rotation every $60$ seconds. Passengers board at the bottom, $2$ m above the ground. Which cosine equation models height $h(t)$ in meters?

**a.** $h(t) = -25\cos\left(\frac{\pi}{30}t\right) + 27$

**b.** $h(t) = 25\cos\left(\frac{\pi}{30}t\right) + 27$

**c.** $h(t) = -25\cos\left(\frac{\pi}{60}t\right) + 2$

**d.** $h(t) = -27\cos\left(\frac{\pi}{30}t\right) + 25$

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Sinusoidal Graphs & Transformations

What is the phase shift of $g(x) = 2\sin\left(4x + \frac{\pi}{2}\right)$?

**a.** $\frac{\pi}{8}$ units to the left

**b.** $\frac{\pi}{8}$ units to the right

**c.** $\frac{\pi}{2}$ units to the left

**d.** $\frac{\pi}{4}$ units to the left

**30** [Intermediate] • Target: Fundamental Trigonometric Identities

Simplify the trigonometric expression $\frac{\tan x}{\sec x}$.

**a.** $\sin x$

**b.** $\cos x$

**c.** $\csc x$

**d.** $\cot x$

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**31** [Intermediate] • Target: Fundamental Trigonometric Identities

Which expression is equivalent to $(1-\cos^2\theta)(1+\cot^2\theta)$ for all defined $\theta$?

**a.** $1$

**b.** $\sin^2\theta$

**c.** $\tan^2\theta$

**d.** $\cos^2\theta$

**32** [Intermediate] • Target: Fundamental Trigonometric Identities

Solve for $\theta \in [0, 2\pi)$: $2\cos^2\theta - \cos\theta - 1 = 0$.

**a.** $\theta = 0, \frac{2\pi}{3}, \frac{4\pi}{3}$

**b.** $\theta = \pi, \frac{\pi}{3}, \frac{5\pi}{3}$

**c.** $\theta = 0, \frac{\pi}{3}, \frac{5\pi}{3}$

**d.** $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$

**33** [Intermediate] • Target: Radian Measure & Unit Circle Coordinates

Write a sinusoidal function with amplitude $5$, period $3\pi$, midline $y = -2$, using $\cos$.

**a.** $f(x) = 5\cos(2/3x) - 1$

**b.** $f(x) = 6\cos(2/3x) - 2$

**c.** $f(x) = 5\cos(4/3x) - 2$

**d.** $f(x) = 5\cos(2/3x) - 2$

**34** [Intermediate] • Target: Exact Values of Trigonometric Functions

Write a sinusoidal function with amplitude $5$, period $4\pi$, midline $y = -1$, using $\cos$.

**a.** $f(x) = 5\cos(1/2x) - 1$

**b.** $f(x) = 5\cos(1x) - 1$

**c.** $f(x) = 6\cos(1/2x) - 1$

**d.** $f(x) = 5\cos(1/2x) + 0$

**35** [Intermediate] • Target: Sinusoidal Graphs & Transformations

Write a sinusoidal function with amplitude $3$, period $4\pi$, midline $y = 2$, using $\sin$.

**a.** $f(x) = 4\sin(1/2x) + 2$

**b.** $f(x) = 3\sin(1/2x) + 3$

**c.** $f(x) = 3\sin(1/2x) + 2$

**d.** $f(x) = 3\sin(1x) + 2$

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**36** [Intermediate] • Target: Fundamental Trigonometric Identities

Write a sinusoidal function with amplitude 3, period $3\pi$, midline $y = 1$, using $\sin$.

**a.** $f(x) = 3\sin(2/3x) + 1$

**b.** $f(x) = 3\sin(4/3x) + 1$

**c.** $f(x) = 3\sin(2/3x) + 2$

**d.** $f(x) = 4\sin(2/3x) + 1$

**37** [Intermediate] • Target: Radian Measure & Unit Circle Coordinates

Write a sinusoidal function with amplitude 2, period $3\pi$, midline $y = 0$, using $\cos$.

**a.** $f(x) = 3\cos(2/3x)$

**b.** $f(x) = 2\cos(2/3x) + 1$

**c.** $f(x) = 2\cos(4/3x)$

**d.** $f(x) = 2\cos(2/3x)$

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Exact Values of Trigonometric Functions • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Write a sinusoidal function with amplitude 5, period $2\pi$, midline $y = -2$, using $\cos$.

**a.** $f(x) = 5\cos(2x) - 2$

**b.** $f(x) = 5\cos(1x) - 2$

**c.** $f(x) = 5\cos(1x) - 1$

**d.** $f(x) = 6\cos(1x) - 2$

**39** [Challenging] • Target: Sinusoidal Graphs & Transformations • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Write a sinusoidal function with amplitude 4, period $2\pi$, midline $y = -3$, using $\cos$.

**a.** $f(x) = 4\cos(1x) - 2$

**b.** $f(x) = 4\cos(2x) - 3$

**c.** $f(x) = 5\cos(1x) - 3$

**d.** $f(x) = 4\cos(1x) - 3$

**40** [Challenging] • Target: Fundamental Trigonometric Identities • Cross-Lesson: 4.01 Inverse & Composite Functions

Write a sinusoidal function with amplitude 2, period $2\pi$, midline $y = 3$, using $\sin$.

**a.** $f(x) = 2\sin(2x) + 3$

**b.** $f(x) = 2\sin(1x) + 4$

**c.** $f(x) = 2\sin(1x) + 3$

**d.** $f(x) = 3\sin(1x) + 3$

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**41** [Challenging] • Target: Radian Measure & Unit Circle Coordinates • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Write a sinusoidal function with amplitude 3, period $\pi$, midline $y = -2$, using $\cos$.

**a.** $f(x) = 4\cos(2x) - 2$

**b.** $f(x) = 3\cos(2x) - 1$

**c.** $f(x) = 3\cos(4x) - 2$

**d.** $f(x) = 3\cos(2x) - 2$

**42** [Challenging] • Target: Exact Values of Trigonometric Functions • Cross-Lesson: 4.03 Parametric Equations & Modeling

Write a sinusoidal function with amplitude 4, period $\pi$, midline $y = -3$, using $\cos$.

**a.** $f(x) = 4\cos(2x) - 3$

**b.** $f(x) = 4\cos(4x) - 3$

**c.** $f(x) = 5\cos(2x) - 3$

**d.** $f(x) = 4\cos(2x) - 2$

**43** [Challenging] • Target: Sinusoidal Graphs & Transformations • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Write a sinusoidal function with amplitude 4, period $\pi$, midline $y = -3$, using $\sin$.

**a.** $f(x) = 4\sin(4x) - 3$

**b.** $f(x) = 4\sin(2x) - 2$

**c.** $f(x) = 5\sin(2x) - 3$

**d.** $f(x) = 4\sin(2x) - 3$

**44** [Challenging] • Target: Fundamental Trigonometric Identities • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Write a sinusoidal function with amplitude 5, period $\pi$, midline $y = -1$, using $\sin$.

**a.** $f(x) = 5\sin(2x) + 0$

**b.** $f(x) = 6\sin(2x) - 1$

**c.** $f(x) = 5\sin(4x) - 1$

**d.** $f(x) = 5\sin(2x) - 1$

**45** [Challenging] • Target: Radian Measure & Unit Circle Coordinates • Cross-Lesson: 1.03 Rates of Change & Concavity

Write a sinusoidal function with amplitude 4, period $\pi$, midline $y = 2$, using $\sin$.

**a.** $f(x) = 5\sin(2x) + 2$

**b.** $f(x) = 4\sin(4x) + 2$

**c.** $f(x) = 4\sin(2x) + 2$

**d.** $f(x) = 4\sin(2x) + 3$

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